SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 - Error Analysis Task

Course Code	CSA17	Course Title	Artificial Intelligence
CO Assessed	CO1 - Apply (BL3)	Assessment	Assessment Tool 3 - Error Analysis Task
Weightage	35% of CIA	Duration / Marks	60 mins / 100 marks

CO1 Statement: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

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Code of Conduct

I _____, _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Contents: Question 1 (Foundations, Agents, Rationality) | Question 2 (Problem Formulation - Missionaries and Cannibals) | Question 3 (BFS / DFS / UCS error traces) | Question 4 (Belief-state search) | Rubrics and Evaluation page.

Grey text in each question = the faulty work given in the question paper. Coloured / normal text = my analysis and answers.

Question 1: Foundations of AI, Intelligent Agents and Rationality - Error Analysis of Concept Notes

Given notes

No.	Statement in the notes
S1	The Turing Test (1950) judges a machine to be intelligent if its internal reasoning always follows the laws of logic. It is therefore the standard example of the "thinking rationally" approach to AI.
S2	The "AI winters", periods of reduced funding and interest, occurred in the 1950s, before the 1956 Dartmouth workshop. That workshop then revived the field and coined the term "artificial intelligence".
S3	A rational agent always achieves the best possible outcome because it knows the actual result of every action in advance. Hence a taxi that picks a route later blocked by an unforeseeable landslide was not rational.
S4	PEAS description of an automated taxi - Performance measure: steering, accelerator, brake, horn. Environment: roads, other traffic, pedestrians, weather. Actuators: safe, fast, legal and comfortable trip. Sensors: cameras, GPS, speedometer.
S5	Chess played without a clock is a partially observable, stochastic, episodic, static and discrete task environment.
S6	A model-based reflex agent keeps an internal state, updated using knowledge of how the world evolves and how its own actions affect it. It can therefore act sensibly even when the current percept does not reveal the full state of the world.

(a) Correct / incorrect statements

Statement	Correct / Incorrect	Erroneous phrase / claim
S1	Incorrect	"judges a machine to be intelligent if its internal reasoning always follows the laws of logic" and "standard example of the 'thinking rationally' approach".
S2	Incorrect	"occurred in the 1950s, before the 1956 Dartmouth workshop" (wrong dating) and "That workshop then revived the field" (it launched the field; nothing existed to revive).
S3	Incorrect	"always achieves the best possible outcome because it knows the actual result of every action in advance" (this describes omniscience) and the conclusion "was not rational".
S4	Incorrect (more than one error)	"Performance measure: steering, accelerator, brake, horn" and "Actuators: safe, fast, legal and comfortable trip" - the two components are swapped. (Environment and Sensors are correct.)
S5	Incorrect (more than one error)	"partially observable" (should be fully observable), "stochastic" (should be deterministic / strategic) and "episodic" (should be sequential). "static" and "discrete" are correct.
S6	Correct	None.

(b) Root cause analysis

Error (statement / part)	Root cause (misconception)
S1 - judged by internal logic; 'thinking rationally'	Confuses two of the four approaches to AI. The Turing Test is a purely behavioural test: a machine passes if a human interrogator cannot tell its written replies from a human's. It therefore belongs to **acting humanly**. "Thinking rationally" is the laws-of-thought / logic approach (Aristotle's syllogisms, logicist tradition) which is concerned with correct inference inside the system, not with imitation of human behaviour.
S2 - AI winters placed before 1956	Timeline confusion. 1956 Dartmouth workshop is the birth of AI; the early years were of great enthusiasm. The winters came *after* the hype failed to deliver - roughly the mid-1970s and the late-1980s to early-1990s (limits of early methods, collapse of the expert-system market). The student reversed cause and effect.
S2 - 'workshop revived the field'	'Revived' presupposes an existing field in decline. The workshop (McCarthy, Minsky, etc.) is where the name 'artificial intelligence' was coined and the field was founded, so it started the field rather than reviving it.
S3 - 'always best outcome / knows actual result'	Confuses **rationality with omniscience** (and perfection). A rational agent maximises *expected* performance given what it has perceived and knows; it cannot know actual outcomes in advance, and unforeseen events do not make a decision irrational.
S3 - 'was not rational'	Judges the decision by its outcome instead of by the information available when the decision was taken. Rationality is assessed through the four factors (performance measure, prior knowledge, available actions, percept sequence), not hindsight.
S4 - performance measure vs actuators swapped	Misunderstanding of PEAS. The **performance measure** is the criterion that defines success (safe, fast, legal, comfortable trip, profit); **actuators** are the means by which the agent acts on the environment (steering, accelerator, brake, horn). The student mixed up the objective with the means.
S5 - 'partially observable'	In chess the whole board, every piece and whose turn it is are visible at every move, so the agent sees the complete state. Confuses the unpredictability of the opponent's choice with hidden state.
S5 - 'stochastic'	Chess has no chance element: the result of a move is fully determined by the state and the action. The opponent makes it multi-agent (strategic), but this is not randomness.

Error (statement / part)	Root cause (misconception)
S5 - 'episodic'	Each move changes the position and affects all later decisions, so experience is not split into independent episodes - it is sequential. Episodic would mean each percept-action pair is independent (e.g. classifying parts on a conveyor).

(c) Corrected statements and justification

Statement	Corrected statement	Justification
S1	The Turing Test (1950) judges a machine to be intelligent if a human interrogator, conversing in writing, cannot distinguish its responses from those of a human. It is the standard example of the "acting humanly" approach.	The test measures external behaviour only and says nothing about the internal reasoning; "thinking rationally" is the logic / laws-of-thought approach.
S2	The "AI winters", periods of reduced funding and interest, occurred *after* the 1956 Dartmouth workshop (roughly mid-1970s and late 1980s to early 1990s). The workshop launched AI as a field and coined the term "artificial intelligence".	Dartmouth 1956 is the founding event; the winters followed the over-optimistic early period.
S3	A rational agent selects the action that is expected to maximise its performance measure, given its percept sequence to date and its built-in knowledge. Rationality is not omniscience, so a taxi that picked a route later blocked by an unforeseeable landslide was still rational.	Rationality depends on expected, not actual, outcomes; the agent cannot perceive or know the future landslide.
S4	Performance measure: safe, fast, legal and comfortable trip. Environment: roads, other traffic, pedestrians, weather. Actuators: steering, accelerator, brake, horn. Sensors: cameras, GPS, speedometer.	Performance measure = what counts as success; actuators = how the agent acts. E and S were already correct.
S5	Chess played without a clock is a fully observable, deterministic (strategic, multi-agent), sequential, static and discrete task environment.	Full board is visible, moves have no random effect, moves affect later positions, the world does not change while the agent deliberates (no clock), and states / moves are finite and distinct.
S6	No change.	A model-based reflex agent maintains an internal state, updated with a model of how the world evolves and how its actions affect it, so it can handle partially observable environments. The statement is accurate.

(d) Application to the self-driving taxi case

(i) Was the choice rational? Yes. Using the four factors that decide rationality:

Factor	Application to the taxi
1. Performance measure	Safe, fast, legal, comfortable trip. The highway promised the best expected value on these criteria.
2. Agent's prior knowledge of the environment	Knows road network, typical highway speeds and risks; nothing in it predicts a random landslide.
3. Actions available	Could choose among routes, speed, braking etc.; choosing the highway was a feasible action.
4. Percept sequence to date	Every available traffic report showed the road to be clear - the landslide was not in any percept.

Given these four factors, the highway route was the action with the highest *expected* performance. The bad outcome came from an event that could not have been perceived, so the agent is rational but not omniscient. (After the landslide happens, the rational response is to perceive it, brake safely and re-plan.)

(ii) Properties that make perfect foresight impossible: the task environment is **partially observable** (the agent cannot sense the slope condition or future events), **stochastic** (outcomes of actions and events like landslides are uncertain) and **dynamic** (the world keeps changing while the agent acts). It is also sequential, continuous and multi-agent, which adds to the uncertainty.

(iii) Simple reflex vs model-based reflex taxi. *Situation:* the taxi wants to change lanes while another car sits in its blind spot (or has just dropped out of the camera view behind a truck). A simple reflex taxi acts on the current percept only ("lane looks empty"), so it changes lanes and collides. A **model-based reflex taxi keeps an internal state**: the last seen positions and speeds of nearby vehicles, a model of how the world evolves (the car keeps moving at about its previous speed) and of how its own actions affect the world (steering moves the taxi sideways). With this it infers that the hidden car is still there and does not change lane.

Question 2: Problem-Solving Agents and Problem Formulation - Error Analysis of a Formulation

Given: Missionaries and Cannibals - 3 missionaries and 3 cannibals on the left bank, boat carries 1 or 2 people and cannot cross empty, boat starts on left bank, missionaries must never be outnumbered by cannibals on either bank (a bank with no missionaries is safe). Objective: move everyone to the right bank with the fewest crossings.

Component	Student's formulation
State representation	(M, C) = number of missionaries and cannibals on the left bank
Initial state	(3, 3)
Actions	Take the boat across with 1M, 2M, 1C, 2C or 1M + 1C
Goal test	M = C
Legality check	A state is legal if C <= M on the left bank (or M = 0 there)
Path cost	Number of people carried in the crossing
Note N1	The agent works in the order: search -> goal formulation -> problem formulation -> execution.
Note N2	The state space contains 16 states, of which 10 are legal.

(a) Correct / incorrect

Component / Note	Correct / Incorrect	What is wrong
State representation	Incorrect	Boat position is missing. (M, C) cannot tell on which bank the boat is, so applicable actions and results cannot be determined, and (2,2) with the boat left is not distinguished from (2,2) with the boat right.
Initial state	Incorrect	(3, 3) is incomplete: the problem says the boat starts on the left bank, so it must be (3, 3, L).
Actions	Correct	1M, 2M, 1C, 2C and 1M+1C are exactly the non-empty loads of at most 2 people (boat is applicable from the side it is on and needs enough people there).
Goal test	Incorrect	M = C is true for the initial state (3,3) and for (2,2), (1,1), (0,0); it does not say that everyone has crossed. Goal = all six on the right bank.
Legality check	Incorrect	Only the left bank is checked. The right bank must satisfy the same rule: with M' = 3 - M and C' = 3 - C, need M' = 0 or C' <= M'.
Path cost	Incorrect	Objective is the fewest crossings, so each crossing should cost 1. Counting people carried optimises a different quantity.
Note N1	Incorrect	Wrong order. A problem-solving agent does: goal formulation -> problem formulation -> search -> execution.
Note N2	Incorrect	Ignores the boat. With boat side, there are 4 x 4 x 2 = 32 states, of which 10 x 2 = 20 are legal. (Also, the student's own legality check would accept 13 (M, C) pairs, not 10.)

(b) Root cause analysis

Error (component / note)	Root cause
State representation and initial state	A state description must contain *all* information needed to decide which actions apply and what they lead to. Omitting the boat position leaves the state description incomplete (the next state is no longer determined by state + action).
Goal test	A goal test must be true only for goal states (and for every goal state). The student's test describes a property (equal numbers) that is not the problem's goal, so it would stop the search at the start state.
Legality check	The safety constraint applies to both banks; the student applied it to one bank only and forgot that moving people changes the right bank as well.
Path cost	Path cost must encode the stated performance measure. The objective was fewest crossings, so cost must be per crossing, not per person.
Note N1	Misunderstanding of the problem-solving agent cycle: the goal limits what is worth searching for, the problem is formulated for that goal, and search needs a formulated problem; only then can execution begin.
Note N2	Counting states from an incomplete representation: omitting a state variable (boat side, 2 values) halves the true state-space size.

(c) Corrected formulation and justification

Component / Note	Corrected formulation	Justification
State representation	(M, C, B): M and C = missionaries and cannibals on the **left** bank (0-3), B = boat side (L or R).	Contains everything needed; the right bank is derived as (3 - M, 3 - C).
Initial state	(3, 3, L)	All on left bank, boat on left.

Component / Note	Corrected formulation	Justification
Actions	No change: move 1M, 2M, 1C, 2C or 1M+1C across, from the boat's current side, provided enough people are on that bank.	Already correct.
Goal test	State = (0, 0, R)	Everyone, and the boat, on the right bank; true only for the goal.
Legality check	Legal if (M = 0 or C <= M) **and** (3 - M = 0 or 3 - C <= 3 - M).	Safety must hold on both banks.
Path cost	Step cost 1 per crossing; path cost = number of crossings.	Matches 'fewest crossings'.
Note N1	Goal formulation -> problem formulation -> search -> execution.	Standard problem-solving agent cycle.
Note N2	32 states in all (4 x 4 x 2), of which 20 are legal.	10 legal (M, C) pairs x 2 boat positions.

(d) Application

(i) Boat loads from the initial state (3, 3, L) (boat moves to R, so the resulting state has B = R):

Boat load	Resulting state (M, C, boat)	Legal?	Reason
1M	(2, 3, R)	No	Left bank has 2M and 3C: missionaries outnumbered.
2M	(1, 3, R)	No	Left bank has 1M and 3C: missionaries outnumbered.
1C	(3, 2, R)	Yes	Left 3M, 2C is safe; right has 0M, 1C (no missionaries, so safe).
2C	(3, 1, R)	Yes	Left 3M, 1C is safe; right has 0M, 2C (safe).
1M + 1C	(2, 2, R)	Yes	Left 2M, 2C is safe; right 1M, 1C is safe.

(ii) State accepted by the student's check but actually illegal: (M, C) = (1, 0), e.g. (1, 0, R). The left bank (1M, 0C) satisfies C <= M, but the right bank has 2M and 3C - missionaries are outnumbered. Another example is (2, 1).

(iii) Optimal solution cost = 11 crossings (step cost 1). One optimal solution:

Crossing	Move	State after (M, C, B)
1	2C go left -> right	(3, 1, R)
2	1C returns	(3, 2, L)
3	2C go left -> right	(3, 0, R)
4	1C returns	(3, 1, L)
5	2M go left -> right	(1, 1, R)
6	1M + 1C return	(2, 2, L)
7	2M go left -> right	(0, 2, R)
8	1C returns	(0, 3, L)
9	2C go left -> right	(0, 1, R)
10	1C returns	(0, 2, L)
11	2C go left -> right	(0, 0, R) = GOAL

(With the student's cost - people carried - the same path would cost 18, which shows why that cost does not represent the number of crossings.)

Question 3: Searching for Solutions and Uninformed Search - Error Analysis of Search Traces

Given: weighted undirected graph, start S, goal G. Graph search with explored set; neighbours expanded alphabetically; goal test when a node leaves the frontier; ties alphabetical.

Edge	Cost	Edge	Cost
S - C	3	B - G	3
S - D	3	D - E	2
A - B	2	D - G	6
A - C	6	E - G	3
B - D	4		

Student's BFS trace (queue initially [S])

Iteration	Node dequeued	Queue after expansion	Explored set
1	S	[C, D]	{S}
2	C	[D, A]	{S, C}
3	D	[B, E, G, A]	{S, C, D}
4	B	[E, G, A]	{S, C, D, B}
5	E	[G, A]	{S, C, D, B, E}
6	G (goal)	[A]	{S, C, D, B, E}

BFS result: path S-D-G, cost 9, 6 nodes removed from the frontier.

Student's DFS trace (stack top on the left)

Iteration	Node popped	Stack after pushing children
1	S	[D, C]
2	D	[G, E, B, C]
3	G (goal)	[E, B, C]

DFS result: path S-D-G, cost 9, 3 nodes removed from the frontier.

Student's UCS trace

Iteration	Node expanded	g(n)	Priority queue after expansion
1	S	0	C:3(S), D:3(S)
2	C	3	D:3(S), A:9(C)
3	D	3	E:5(D), B:7(D), A:9(C), G:9(D)
4	E	5	B:7(D), A:9(C), G:9(D)
5	B	7	A:9(C), G:9(D)
6	A	9	G:9(D)
7	G (goal)	9	-

UCS result: path S-D-G, cost 9, 7 nodes removed from the frontier.

No.	Student's remark
R1	An explored set is not needed in BFS because BFS never revisits a node, so the Explored Set column is unnecessary.
R2	All three strategies returned S-D-G with cost 9, so all three are optimal and the traces are confirmed correct.
R3	DFS removed the fewest nodes from the frontier (3), so DFS is the best strategy for this graph.

(a) Errors found

No.	Where (trace / iteration / remark)	What is wrong
1	BFS, iteration 3	Queue after expanding D written as [B, E, G, A]. A was already in the queue (added at iteration 2), ahead of the new nodes, so the correct queue is **[A, B, E, G]**. (Carried forward, counted once: iteration 4 should dequeue A, not B; the BFS result should be 7 nodes removed, not 6.)
2	DFS, iteration 1	Stack [D, C]: children pushed in reverse. Alphabetically first child C must be on top: **[C, D]**. (Carried forward: D popped next, path S-D-G, cost 9, 3 nodes - all wrong.)
3	DFS, iteration 2	The same push-order rule is violated again in a fresh expansion: children of D (B, E, G) pushed as [G, E, B, C] instead of [B, E, G, C] (B must be on top).
4	UCS, iteration 4	When E (g = 5) is expanded, G is reached with cost 5 + 3 = 8, which is cheaper than G:9(D) in the queue. The entry must be replaced by **G:8(E)**; the student left G:9(D). (Carried forward: A expanded at iteration 6, goal removed at 9, 7 nodes - wrong.)

No.	Where (trace / iteration / remark)	What is wrong
5	Remark R1	BFS in graph search does need an explored set. In an undirected graph, every child has its parent as a neighbour (e.g. S appears again when C and D are expanded), so without the explored set S would be re-added.
6	Remark R2	Equal cost does not prove optimality, and the traces are not 'confirmed correct'. BFS and DFS ignore edge costs and are not cost-optimal; only UCS is. The correct costs are BFS 9, DFS 14, UCS 8, so the three were not equal.
7	Remark R3	Fewest nodes removed is not a measure of the best strategy. Corrected DFS removes 5 nodes but returns a path of cost 14, the worst of the three; the best (cheapest) path is found by UCS.

(b) Root cause analysis

No.	Root cause (rule or concept violated)
1	BFS uses a FIFO queue: nodes already in the queue stay ahead of newly generated nodes, and new children are appended at the tail. The student appended the older node A after the new children, breaking FIFO order, so the wrong node was dequeued in iteration 4 and the node count was wrong.
2	DFS convention stated in the question: push children so that the alphabetically first child is popped first. Pushing C then D (D on top) reversed the rule; D was explored first, which changed the whole search order and result.
3	Same push-order rule mis-applied again; the student pushed alphabetical order from the bottom instead of making the alphabetically first node the top of the stack.
4	UCS must update a frontier entry whenever a cheaper path to that node is found (the 'replace' rule). The student only appended new nodes and did not compare the new path cost to the existing G:9(D). Missing this relaxation also made UCS return a non-optimal cost.
5	Misunderstanding of graph search vs tree search: an explored set (or frontier check) is what prevents repeated states, particularly in undirected graphs with back edges. BFS avoids loops only because of this check.
6	Confuses 'same answer' with 'correct / optimal'. Optimality depends on the algorithm's guarantee: BFS is optimal only for uniform step cost, DFS is not optimal, UCS is optimal for non-negative costs. Here the traces were also wrong, so equality was an artefact of the errors.
7	Wrong evaluation criterion: strategies are compared by completeness, optimality, time and space together; node count alone ignores solution quality (path cost).

(c) Corrected traces and justification

(i) BFS - queue after iteration 3: **[A, B, E, G]**. Order of dequeued nodes: **S, C, D, A, B, E, G**. Full corrected trace:

Iteration	Node dequeued	Queue after expansion	Explored set
1	S	[C, D]	{S}
2	C	[D, A]	{S, C}
3	D	[A, B, E, G]	{S, C, D}
4	A	[B, E, G]	{S, C, D, A}
5	B	[E, G]	{S, C, D, A, B}
6	E	[G]	{S, C, D, A, B, E}
7	G (goal)	[ ]	goal test succeeds on removal

(ii) DFS - corrected trace (stack top on the left):

Iteration	Node popped	Stack after pushing children
1	S	[C, D]
2	C	[A, D] (S explored)
3	A	[B, D] (C explored)
4	B	[G, D] (A explored; D already on stack, not pushed again)
5	G (goal)	[D] goal test succeeds on pop

(iii) UCS - corrected trace:

Iteration	Node expanded	g(n)	Priority queue after expansion
1	S	0	C:3(S), D:3(S)
2	C	3	D:3(S), A:9(C)
3	D	3	E:5(D), B:7(D), A:9(C), G:9(D)
4	E	5	B:7(D), G:8(E), A:9(C) (G replaced: 5 + 3 = 8 < 9)
5	B	7	G:8(E), A:9(C) (A via B = 9 not cheaper; G via B = 10 not cheaper)

Iteration	Node expanded	g(n)	Priority queue after expansion
6	G (goal)	8	A:9(C) remaining - goal test succeeds on removal

Justification of the corrections. BFS: FIFO order keeps A in front of B, E, G, so A is dequeued at iteration 4 and G is removed at iteration 7 with path S-D-G. DFS: pushing C on top makes C the first child explored; the search goes S, C, A, B, G; D stays on the stack and is never expanded. UCS: relaxing G through E gives g = 8 which is smaller than 9, so G leaves the queue before A and the cheapest path S-D-E-G (3 + 2 + 3 = 8) is found.

(d) Application

(i) Comparison table

Strategy	Path found	Path cost	Nodes removed from frontier	Optimal here?
BFS	S - D - G	9	7	No (9 > 8; shallowest, but not cheapest)
DFS	S - C - A - B - G	14	5	No
UCS	S - D - E - G	8	6	Yes

(ii) Why an explored set is needed (BFS iteration 3). At iteration 3, D is dequeued and its neighbours are B, E, G and S. S is the start node that has already been expanded (explored set {S, C, D}); without the explored set it would be added to the queue again: [A, B, E, G, **S**], and S would be expanded again, re-generating C and D, and so on. In an undirected graph every edge can be traversed backwards, so repeated states appear all the time (C also generated S at iteration 2). The explored set rejects S, so only the genuinely new nodes B, E, G are added, which keeps the search finite and avoids duplicated work.

Question 4: Search with Partial Information - Error Analysis of Belief-State Computation

Given: two-square vacuum world. States (Agent, Left, Right): S1 (L,D,D), S2 (R,D,D), S3 (L,D,C), S4 (R,D,C), S5 (L,C,D), S6 (R,C,D), S7 (L,C,C), S8 (R,C,C). Goal: both squares clean (S7, S8). Actions Left, Right, Suck.
Task A (sensorless), plan [Suck, Right, Suck], student's table:

Step	Action	Belief state after action
0 (start)		{S1, S3, S5, S7}
1	Suck	{S5, S7}
2	Right	{S6, S8}
3	Suck	{S6, S8}

Task B (local sensor), student's table:

Step	Action	After action (predicted)	Percept	After percept (updated)
0 (start)		{S1,...,S8}	[Left, Dirty]	{S1, S2, S3, S6}
1	Suck	{S4, S5, S7, S8}	[Left, Clean]	{S5, S7}
2	Right	{S6, S7}	[Right, Dirty]	{S6}

No.	Student's conclusion
C1	Task A: the final belief state still contains two states, so the agent can never know where it is. No plan can guarantee the goal when the agent has no sensors; hence sensorless problems cannot be solved.
C2	Task B: the belief state is now the singleton {S6}, so the agent knows its location and the dirt status, and the goal has been reached.
C3	With 8 physical states there are only 8 belief states (one per physical state), so searching over belief states is no harder than searching over physical states.

(a) Errors found

No.	Where (task / step / conclusion)	What is wrong
1	Task A, step 3 (Suck)	Belief {S6, S8} is unchanged. Suck in S6 (R,C,D) cleans the Right square and gives S8 (R,C,C); S8 stays S8. Correct belief: **{S8}**.
2	Task B, step 0 (percept [Left, Dirty])	Updated belief {S1, S2, S3, S6} contains S2 and S6 (agent on the Right) - inconsistent with the percept. Only states with agent at Left and Left square dirty remain: **{S1, S3}**. (Step 1 predicted {S4, S5, S7, S8} is just carried forward from this error, counted once.)
3	Task B, step 2 (action Right)	Predicted {S6, S7}: S7 (L,C,C) was left in place. Right moves the agent to the Right square: S5 -> S6 and S7 -> S8. Correct predicted belief: **{S6, S8}**.
4	Conclusion C1 - 'final belief has two states so the agent can never know where it is'	Wrong premise (the real final belief is the single state {S8}) and wrong reasoning: even {S6, S8} has the agent on the Right in both states, so location is known; only dirt was uncertain.
5	Conclusion C1 - 'sensorless problems cannot be solved'	Sensorless problems are solvable: the plan [Suck, Right, Suck] ends in {S8}, where every state is a goal state, so it is a solution.
6	Conclusion C2	{S6} = (R, C, D): the Right square is still dirty, so S6 is not a goal state. Knowing the exact state does not mean the goal is reached; one more Suck is needed.
7	Conclusion C3	A belief state is a *set* of physical states, so the number of belief states is $2^8 = 256$ (255 non-empty), not 8; belief-state search is much harder.

(b) Root cause analysis

No.	Root cause (concept or rule misapplied)
1	Prediction step of belief update: the action must be applied to *every* state in the belief state, b' = union of RESULT(s, a). The student applied Suck to S8 only and left S6 unchanged, forgetting that Suck cleans the current (Right) square.
2	Percept-update step: after the prediction, keep only the states consistent with the percept. [Left, Dirty] means location = Left *and* the current (Left) square is dirty. The student matched loosely (any dirt) and included states with the agent on the Right.
3	Prediction step again: Right always puts the agent on the Right square, so every state in the predicted belief must have the agent at R. S7 (L,C,C) cannot stay as S7; it becomes S8.
4	Confuses uncertainty about the dirt with uncertainty about location, and builds on error 1. A belief state can contain several states yet still fix the agent's location.
5	Sensorless problems are solved by **coercion**: actions shrink the belief state without any percept. The solution is a fixed action sequence whose final belief state contains only goal states.
6	Goal test for belief states: the goal is reached only if *all* states in the belief are goal states. Knowing the state (singleton belief) is a different thing from being in a goal state.

No.	Root cause (concept or rule misapplied)
7	Belief states are subsets of the physical state space (power set), not single states; the student equated belief states with physical states.

(c) Corrected tables and conclusions

(i) Task A - corrected belief states

Step	Action	Belief state after action
0 (start)		{S1, S3, S5, S7}
1	Suck	{S5, S7} (S1 -> S5, S3 -> S7, S5 and S7 unchanged)
2	Right	{S6, S8} (S5 -> S6, S7 -> S8)
3	Suck	**{S8}** (S6 -> S8, S8 unchanged) - goal state

(ii) Task B - corrected belief states

Step	Action	After action (predicted)	Percept	After percept (updated)
0 (start)		{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	**{S1, S3}**
1	Suck	**{S5, S7}** (S1 -> S5, S3 -> S7)	[Left, Clean]	{S5, S7}
2	Right	**{S6, S8}** (S5 -> S6, S7 -> S8)	[Right, Dirty]	{S6} (S8 is clean, removed)

(iii) Corrected conclusions

	Corrected conclusion and justification
C1	Task A: after the plan [Suck, Right, Suck] the belief state is the singleton {S8}, which is a goal state. Sensorless problems *can* be solved: the agent reaches the goal from any state in its initial belief using coercion, without sensors. Even in the intermediate belief {S6, S8} the agent's location (Right) was known; only the dirt status was uncertain.
C2	Task B: after the last percept the belief state is the singleton {S6}, so the agent knows it is on the Right square with Right dirty and Left clean - but this is **not** a goal state, so the goal has not yet been reached. One more action, Suck, gives {S8}, which is the goal.

(d) Application

(i) Number of belief states. With 8 physical states there are 2^8 = **256** possible belief states (every subset; 255 if the empty set is excluded). Belief-state search is harder because the belief-state space grows exponentially (N physical states -> 2^N belief states), each belief state may hold many physical states that all have to be updated by every action and percept, and so the search space, memory and branching to be handled are far larger than for the 8 physical states.

(ii) Plan [Right, Suck] from {S1, S3, S5, S7}

Step	Action	Belief state after action
0 (start)		{S1, S3, S5, S7}
1	Right	{S2, S4, S6, S8} (S1 -> S2, S3 -> S4, S5 -> S6, S7 -> S8)
2	Suck	{S4, S8} (S2 -> S4, S4 -> S4, S6 -> S8, S8 -> S8)

Solution? No. The final belief {S4, S8} contains S4 = (R, D, C), in which the Left square is still dirty, so not all states are goal states. The plan never cleans the Left square; the plan [Suck, Right, Suck] from Task A is a solution but [Right, Suck] is not.

Rubrics for Error Analysis

Criteria	Wt (%)	Level 4 - Exemplary (4)	Level 3 - Proficient (3)	Level 2 - Developing (2)	Level 1 - Beginning (1)
Error Identification	20	Accurately identifies all errors with clear understanding of issues	Identifies most errors with minor omissions	Identifies limited errors; some are missed	Fails to identify key errors
Root Cause Analysis	30	Clearly explains underlying causes with strong technical reasoning	Explains causes with moderate clarity	Partial explanation; weak reasoning	Unable to identify causes of errors
Correction & Justification	25	Provides correct solutions with strong justification and logic	Provides correct corrections with some justification	Corrections are partially correct or weakly justified	Incorrect or no corrections provided
Application of Concepts	15	Effectively applies relevant concepts to analyze and resolve errors	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts
Clarity & Presentation	10	Presents analysis clearly, logically, and systematically	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Evaluation Page

Student: Afrose A | Reg. No.: 1924260309

Q. No. / Criteria	Error Identification	Root Cause Analysis	Correction & Justification	Application of Concepts	Clarity & Presentation	Sub Total
1						
2						
3						
4						
Total						
%						

Faculty In-charge	Course Coordinator	Program Director
Signature: _____ Date: _____	Signature: _____ Date: _____	Signature: _____ Date: _____